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Roll No. ....

Degree: B-Tech (ECAM)

Semester: III — Course work

**MID-SEMESTER EXAMINATION, September, 2025**

**Course Title: DIGITAL CIRCUITS & SYSTEMS**

**Course Code: EAEPC305**

Duration: 1Hr 30 Minutes

Max. Marks: 20

*Note: - Attempt all questions in the given order only. Missing data/information (if any), may be suitably assumed & mentioned in the answer.*

| Q. No. | Question                                                                                                                                 | Marks | CO    |
|--------|------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
| 1a     | Find the Octal, BCD, Excess-3 and Hexadecimal equivalents of $(23)_{10}$                                                                 | 2     | I     |
| 1b     | If $x = (14.75)_{10}$ and $y = (9.25)_{10}$ , convert x and y to binary and find their difference $(x - y)$ using 2's complement method. | 2     | I     |
| 2a     | If $p = 724$ and $q = 539$ , find the BCD sum of p and q.                                                                                | 2     | I     |
| 2b     | Using truth table, prove that complement of $(\bar{A}.B + A.\bar{B})$ is equal to $(\bar{A}.\bar{B} + A.B)$                              | 2     | I     |
| 3a     | Design a 3 bit Binary to Gray code converter.                                                                                            | 2     | I, II |
| 3b     | Design a Half Subtractor, implement using only universal gates.                                                                          | 2     | II    |
| 4a     | Minimize using k-map:<br>$F(A,B,C,D) = \sum_{\min} (2, 8, 12, 14) + d(4, 5, 6, 10)$                                                      | 2     | II    |
| 4b     | Draw and explain the logic diagram of a single digit BCD adder.                                                                          | 2     | II    |
| 5a     | Design an 8:1 multiplexer, using two 4:1 multiplexers.                                                                                   | 2     | II    |
| 5b     | Implement using 8:1 Multiplexer:<br>$F(w, x, y, z) = \sum_{\min} (0, 1, 6, 7, 9, 12, 13, 14)$                                            | 2     | II    |